

Tanvir Rahman Anik

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PROFILE SUMMARY

Machine Learning Engineer with extensive experience in Python-based software development, building scalable web applications and AI-powered systems. Proficient in Django, FastAPI, and Flask for developing API-driven platforms and enterprise applications. Strong foundation in object-oriented programming (OOP), system design, and relational databases. Experienced in integrating machine learning and data-driven automation into production software systems.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer May 2025 – Present
Interactive Cares, Dhaka, Bangladesh

- Engineered an enterprise-grade **conversational AI interview platform** with domain-driven architecture, multi-modal processing, OpenAI+Gemini integration, and Redis caching, reducing candidate screening time by 70%
- Developed a modern, responsive frontend for AI-Interviewer using **Next.js** App Router with server-side rendering (SSR), implementing protected route middleware, role-based authentication (candidate/HR/guest), and real-time interview interfaces with WebRTC video recording and state management using Zustand and React Query.
- Built a scalable **CV screening platform** leveraging **FastAPI microservices**, OpenAI Assistant APIs with rate limiting controls, and intelligent batch processing, automatically scoring and ranking 500+ candidates while reducing manual review time by **80%**.
- Implemented a **job recommendation and resume optimization module** with a **Flask-based backend**, retrieving the most relevant job matches for each candidate and generating **embedding-based feedback** to improve semantic alignment between CV content and job descriptions.

Machine Learning Engineer July 2023 – April 2025
Bista Solutions Inc., Dhaka, Bangladesh

- Fine-tuned **Meta-LLaMA 3.1 (8B)** using QLoRA with 4-bit quantization, achieving **85.48% exact match accuracy** on text-to-SQL generation for ERP databases; demonstrating expertise in high-performance computation and large model optimization.
- Architected and deployed **Retrieval-Augmented Generation (RAG) pipelines** using Django, LangChain, and ChromaDB, integrating embeddings, vector databases, and retrieval workflows to automate ERP tasks such as inventory, order, and finance management.
- Built **Acumen Helpbot**, a document-based RAG assistant enabling seamless PDF interactions and information retrieval through advanced prompt engineering and NLP optimization.
- Developed and integrated **LSTM-based forecasting modules** ($R^2 = 0.87$) for sales trend prediction and business analytics within Odoo's Copilot dashboards, improving inventory planning and decision-making efficiency.
- Executed end-to-end **data extraction and automation workflows** for PDF and image invoices including annotation, preprocessing, model training, and evaluation; ensuring robust vendor data processing pipelines.

TECHNICAL SKILLS

Programming Languages	Python, JavaScript, SQL, C, C++ , Java, HTML, CSS.
Frameworks	Django, FastAPI, Flask, Next.js, React.
Software Development	Object-Oriented Programming (OOP), REST API Development, Microservices Architecture, System Design, Asynchronous Processing.
Machine Learning & AI	Supervised & Unsupervised Learning, Regression, Classification, Clustering, Ensemble Learning, CNN, DNN, Transfer Learning, Transformers.
LLM & AI Tools	LangChain, HuggingFace Transformers, OpenAI Assistants API, Gemini API, Retrieval-Augmented Generation (RAG), Vector Databases.
Data & ML Libraries	TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Pandas, NumPy.
Databases	MySQL, PostgreSQL, MongoDB, Oracle.
Cloud & DevOps	Git, GitHub, Docker, CI/CD Pipelines, Linux, AWS.

EDUCATION

Ahsanullah University of Science and Technology April 2018 – January 2023
Bachelor of Science in Computer Science and Engineering **CGPA: 3.378/4.00**

SELECTED PROJECTS

AI-Interviewer  | *Next.js, TypeScript, FastAPI, OpenAI+Gemini APIs, Redis, MySQL, Docker*

Full-stack AI-powered interview platform using Next.js, TypeScript, and Tailwind CSS on the frontend, integrated with FastAPI microservices backend, delivering an enterprise-grade solution with domain-driven architecture and multi-modal processing via OpenAI+Gemini APIs.

CV-Sorter  | *FastAPI, Python, OpenAI APIs, MySQL, AWS S3*

Enterprise-grade AI-powered CV sorting and candidate selection platform built with FastAPI microservices architecture, featuring async processing, batch operations, and OpenAI integration for intelligent CV analysis and ranking with real-time job tracking and scalable HR analytics.

NewsDocument-Classifer  | *PyTorch, Bi-LSTM, GloVe, Multi-Head Attention, Scikit-learn*

Hierarchical Attention Network for multi-class text classification with dual-stage attention (word-level filtering + document level cross-attention), residual connections, and gating mechanism, achieving 74.22% accuracy on 20 Newsgroups benchmark with extensive attention analysis and error diagnostics.

2D-to-3D Model Generator  | *Python, PyTorch, Transformers, Computer Vision*

A machine learning application that converts 2D images into 3D models using Stability AI's TripoSR transformer architecture, implementing automated 3D mesh generation with OpenCV image processing, Trimesh geometry handling, and GLB/OBJ export functionality for web-compatible 3D visualization.

AR Virtual Hat Try-On Platform  | *Next.js, React Three Fiber, FastAPI, MediaPipe*

A real-time augmented reality web application featuring face detection and 3D hat overlay using MediaPipe's landmark face mesh, implementing WebSocket communication between Next.js frontend and Python FastAPI backend for low-latency frame processing and Three.js 3D rendering with GLB model support.

LLM Fine-tuning for Text-to-SQL Generation  | *Meta-LLaMA, HuggingFace*

Fine-tuned Meta-LLaMA 3.1 (8B) using QLoRA with 4-bit quantization, achieving **85.48%** exact match accuracy on SQL query generation.

Acumen Helpbot  | *Django, ChromaDB, Sentence Transformers*

A RAG-based chatbot with vector search, embeddings-based retrieval, and a latency-optimized response pipeline.

Odoo-Copilot  | *Django, OpenAI Assistants API, LangChain, ChromaDB*

An agentic chatbot for Odoo ERP, enabling natural-language queries, real-time analytics, and workflow automation, leveraging multiple LLMs, LangChain, and vector-based retrieval for enhanced productivity and decision-making.

Knee Osteoarthritis Severity Prediction  | *Python, TensorFlow, Keras, OpenCV*

Deep learning system for KOA severity classification from knee X-rays using pre-trained models and a stacked ensemble, with Grad-CAM, Score-CAM, and LIME for explainability, ensuring compliance with data confidentiality in medical imaging datasets and delivering clinically relevant performance.

Tech Review Platform  | *C#, ASP.NET MVC, JavaScript, SQL Server*

A web application for users to share feedback and reviews on tech products, implementing MVC architecture, role-based authentication, and an interactive front-end for enhanced user experience.

TEACHING EXPERIENCE

Instructor

2018 – 2022

Aid Academy, Science Coaching Center, Dhaka

- Delivered structured lectures in biology and chemistry to high school and college students.
- Provided personal academic assistance and guidance to students preparing for board exams.
- Designed curricula, created exam materials, conducted assessments, and evaluated student performance.

HONORS AND PROFESSIONAL ACTIVITIES

- Second runner-up, Software Project Showcase, AUST CSE Week 2022
- Participated in AUST Programming Contests (2018-2021), Soccer Bot & Line Follower Robot competitions
- Organizing committee member for departmental events, member of AUST Innovation & Design Club (2021-2022)

REFERENCES

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